

MFCR-SleepNet: A hard sample mining-based multi-scale feature contrastive representation learning model for sleep stage classification

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HIGHLIGHTS

- Hard sample mining with dynamic weighting addresses class imbalance and improves N1 stage recognition.
- Supervised contrastive learning optimizes feature space to enhance accuracy on imbalanced datasets.
- A new Boundary Error Rate metric is proposed to assess classification performance at sleep stage transitions.

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ABSTRACT

Automatic but nevertheless accurate sleep stage classification from EEG data is crucial in clinical routine for diagnosing sleep disorders and evaluating sleep quality. Here, we propose a method addressing major problems with this kind of data. Typically, such a dataset is highly imbalanced. Specifically, the “N1” class consists only of very few samples and as a result, classification performance is very low here. From a methodological point of view, the temporal relationships within time series are not sufficiently utilized, which leads to higher classification errors on all “boundary” samples such as N1. In this paper, we introduce the Multi-scale Feature Contrastive Representation Learning Sleep Network (MFCR-SleepNet), which addresses both problems and requires only single-channel EEG data to achieve better results for classifying sleep stages automatically. The model comprises a supervised contrastive representation learning (CRL) module and a pyramid temporal context learning (PTCL) module. The CRL module, combined with hard sample mining, optimizes the representation capability for boundary transitions and minority class samples. The PTCL module captures dynamic features of EEG signals across short-, medium-, and long-term time scales, enabling better excavation of temporal relationships between sleep stages. To evaluate the model's classification ability for boundary samples during sleep stage transitions, we additionally introduce the “Boundary Error Rate” (BER) as a more appropriate metric. Our method achieves accuracies of 85.9 %, 87.0 %, and 87.9 % on the SleepEDF-78, MASS, and SHHS public datasets, respectively. Our experimental results further demonstrate that MFCR-SleepNet performs well across different datasets. Moreover, the BER values on the three datasets are 0.2174, 0.1932, and 0.2493, respectively, indicating that the model significantly improves the classification precision for boundary and minority class samples, effectively addressing the class imbalance problem.

1. Introduction

Sleep is one of the most essential physiological activities for humans. However, sleep disorders and insufficient sleep are widespread globally, posing serious public health challenges. Adequate sleep supports

the normal functioning of various physiological processes in the human body [1] and enhances learning, memory, attention, mood, and decision-making processes [2]. Therefore, improving sleep health is a key strategy for mitigating and preventing symptoms of mental illnesses

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and neurodegenerative diseases. Sleep stage classification, which identifies critical sleep stages for the diagnosis and treatment of sleep disorders, plays a significant role in medical research and practice [3]. For example, the percentage of N1 sleep relative to total sleep may serve as an effective indicator in the assessment of hypersomnolence [4] or insomnia [5]. Traditionally, sleep experts determine sleep stages through polysomnography (PSG), a method that integrates multiple physiological signals, including electroencephalograms (EEG), electrooculograms (EOG), electromyograms (EMG), and electrocardiograms (ECG) [6]. However, the large volume of data generated by PSG makes manual classification a complex and time-consuming process. In contrast, single-channel EEG-based sleep staging has emerged as an attractive alternative due to its simplicity, low cost, and ease of use [7,8]. Sleep experts typically segment PSG or single-channel EEG data into 30-second epochs based on sleep manuals, such as Rechtschaffen and Kales (R&K) [9] and the American Academy of Sleep Medicine (AASM) [10], and then manually inspect and assign sleep stages to each epoch. This process is labor-intensive [11] and subject to inter-expert variability. Moreover, the accuracy of manual classification relies on the expertise of technicians, which is particularly challenging in resource-limited settings. Benefiting from the widespread application of deep learning methods in health monitoring [12–14], automated sleep stage classification approaches based on machine learning or deep learning have attracted considerable research interest due to their high efficiency and accuracy. These approaches not only reduce manual intervention significantly but also improve the efficiency and accuracy of sleep staging.

Traditional machine learning methods [15–17] typically rely on handcrafted features for classification. This process is not only time-consuming but also labor-intensive. Furthermore, handcrafted features often prove suboptimal or even unreliable when applied to new datasets. To address these limitations, deep learning-based approaches for automatic sleep stage classification [18–23] have gradually replaced traditional machine learning methods. These approaches automatically extract effective features from data, eliminating the tedious process of manual feature extraction and demonstrating better performance across multiple datasets by enhancing generalization capabilities. Based on the underlying architecture, sleep stage classification models can be broadly categorized into convolutional neural network (CNN)-based and recurrent neural network (RNN)-based models, along with various combinations and variants. CNN-based methods [24] directly extract features from raw data for sleep stage classification. For instance, Tsinalis et al. [25] developed a CNN-based automatic sleep staging method using single-channel EEG signals, achieving satisfactory classification performance without relying on prior knowledge. Phan et al. [26] proposed an efficient CNN model based on time–frequency image features, which utilized convolution kernels of different sizes to learn multi-scale representations for sleep stage classification. However, due to the temporal nature of sleep EEG signals, CNNs may struggle to capture inherent temporal dependencies effectively. In contrast, methods based on the RNN architecture, such as LSTM and GRU, are more adept at capturing the sequential relationships between sleep stages. For example, Phan et al. [27] proposed the SeqSleepNet architecture based on RNN to handle sequential signals, while Pham et al. [28] employed LSTM to tackle the challenge of recognizing temporal patterns. Consequently, researchers have integrated CNN and RNN modules, where CNNs extract time-invariant features and RNNs learn transitional patterns between different sleep stages, thereby improving classification accuracy [29]. However, the recurrent structure of RNNs results in lower computational efficiency, higher model complexity, and challenges such as gradient explosion when handling long sequences. To overcome these limitations, researchers have explored the use of Transformer models [30], which can more efficiently model global dependencies in time series data. Additionally, 3D-CNNs [31] and methods that replace convolutional layers with wavelet filters [32] have also been introduced.

In recent years, contrastive representation learning has garnered extensive application in EEG classification tasks, owing to its ability to reveal rich latent information in data. Representation learning facilitates more effective extraction of features from EEG data [33]. Furthermore, contrastive learning principles effectively differentiate and characterize features across diverse sleep stages [34], leading to accurate and efficient automatic classification of sleep stages. For instance, Eldele et al. [34] proposed a novel time-series representation learning framework, TS-TCC (Time and Context Contrast), specifically designed for time-series signals such as EEG. Leveraging a self-supervised contrastive learning strategy, this framework achieved robust classification performance on multiple time-series datasets, including SleepEDF. Similarly, Jiang et al. [35] developed a self-supervised contrastive learning approach tailored for EEG-based automatic sleep scoring. Ye et al. [36] proposed a self-supervised learning framework for automatic sleep stage classification using multi-view EEG representations derived from both raw signals and spectrograms.

Although these methods have shown promising performance in capturing and analyzing EEG features for sleep staging and partially mitigating the issue of limited annotated PSG data, they still present the following challenges. Firstly, as mentioned earlier, CNN-based methods [25,26] face difficulties in capturing temporal dependencies and thus fail to adequately exploit contextual information across different sleep stages, particularly when handling boundary stage samples. Conversely, RNN-based methods [27], while capable of processing long temporal sequences, suffer from low computational efficiency and high model complexity, which renders them susceptible to gradient explosion issues. Although LSTM partially alleviates this issue, it still suffers from similar limitations. Secondly, current contrastive learning methods do not fully leverage the available annotated data to address the class imbalance problem in sleep stage classification [37]. This leads to suboptimal classification performance for the N1 stage, ultimately constraining overall classification accuracy. Addressing these challenges is essential for enhancing the accuracy of sleep staging.

This paper introduces the Multiscale Feature Contrastive Representation Learning Sleep Network (MFCR-SleepNet), a model for sleep stage classification leveraging single-channel EEG. MFCR-SleepNet comprises two core modules: a supervised contrastive representation learning (CRL) module and a pyramid temporal context learning (PTCL) module. The CRL module generates positive and negative sample pairs via data augmentation and utilizes a momentum encoder along with a projection network to map features into an embedding space for contrastive representation learning. A hard sample mining strategy is integrated to enhance classification performance for minority sleep stage classes. The PTCL module integrates a multiscale spatial pyramid pooling network (XSPP-CNN) with a self-attention mechanism. XSPP-CNN processes temporal information at multiple scales, while the self-attention mechanism captures long-range dependencies, ensuring comprehensive representations of sleep signals and effectively extracting multidimensional EEG features. The main contributions of this study are as follows:

- 1) We designed a contrastive representation learning module (CRL) that integrates supervised contrastive learning with a momentum encoder to optimize intra-class similarity and inter-class separability, leveraging annotated PSG data to enhance sleep stage classification accuracy.
- 2) We proposed a hard sample mining strategy to tackle boundary ambiguity and class imbalance. Combined with a sample reweighting mechanism, this strategy strengthens recognition of minority classes and boosts classification performance for challenging stages such as N1.
- 3) We developed a pyramid temporal context learning module (PTCL) that integrates a multiscale spatial pyramid pooling

network (XSPP-CNN) with a self-attention mechanism. This module captures multiscale temporal dynamics in EEG signals, enhancing the model's capacity to analyze complex sleep patterns and boundary-stage samples.

- 4) We introduced a novel metric, Boundary Error Rate (BER), to evaluate the network's effectiveness in addressing boundary ambiguity. BER integrates temporal error (TE) and phase consistency (PC) to measure classification accuracy during sleep stage transitions, providing a comprehensive assessment of the model's ability to resolve boundary fuzziness.

The rest of the paper is organized as follows: Section 2 provides a comprehensive description of the architecture and key components of our proposed model. In Section 3, we detail the datasets, preprocessing methods, experimental setup, evaluation metrics, and baseline networks utilized in this study. Section 4 presents the performance of the proposed model in sleep staging tasks across various datasets, with a particular focus on boundary misclassification rates compared to baseline networks. Additionally, this section includes an ablation study of MFCR-SleepNet to analyze its contributions. Finally, the conclusion of the paper is summarized in Section 5.

2. Methods

This section introduces our network model and methodology. Fig. 1 presents the overall structure of the Multiscale Feature Contrastive Representation Learning Sleep Network (MFCR-SleepNet), which encompasses both contrastive representation learning and pyramid temporal context learning. The input to the network is the raw EEG signal from the Fpz-Cz channel, denoted as $S = \{s_1, s_2, \dots, s_L\}$, where L is the total number of samples in the signal.

2.1. Contrastive representation learning

This module comprises data augmentation, a momentum encoder, a projection network, and a hard sample mining component. We utilize a contrastive representation learning strategy to extract deep features from individual EEG epochs. Under the framework of supervised contrastive learning, the model learns to bring samples from the same class closer while pushing samples from different classes farther apart, thereby enhancing its discriminative capability. Our goal is to learn an embedding space in which EEG epochs belonging to the same sleep stage exhibit similar representations, whereas epochs from different stages differ markedly. In this process, we introduce a hard sample mining approach for samples with low classification accuracy and samples that lie near boundaries to further improve performance.

1) Data Augmentation: To effectively implement contrastive representation learning, our data augmentation strategy is specifically designed to generate reliable positive and negative sample pairs. Positive samples are produced through semantically invariant transformations applied to the same sleep stage, while negative samples are selected from different sleep stages. Drawing from Lee et al. [23], we applied six types of transformations to individual EEG epochs—amplitude scaling, time shifting, amplitude offset, zero-masking, additive Gaussian noise, and a band-stop filter with a probability of 0.5—to enhance the diversity of both positive and negative sample pairs. This augmentation increases the volume of training data and enhances its diversity, enabling the network to generalize better to new, unseen data. It is defined as follows:

$$\bar{s} = \text{Aug}(s) \quad (1)$$

Where $\bar{s}$ denotes an alternative yet semantically invariant representation of a single EEG epoch s . Specifically, for a set of randomly sampled EEG

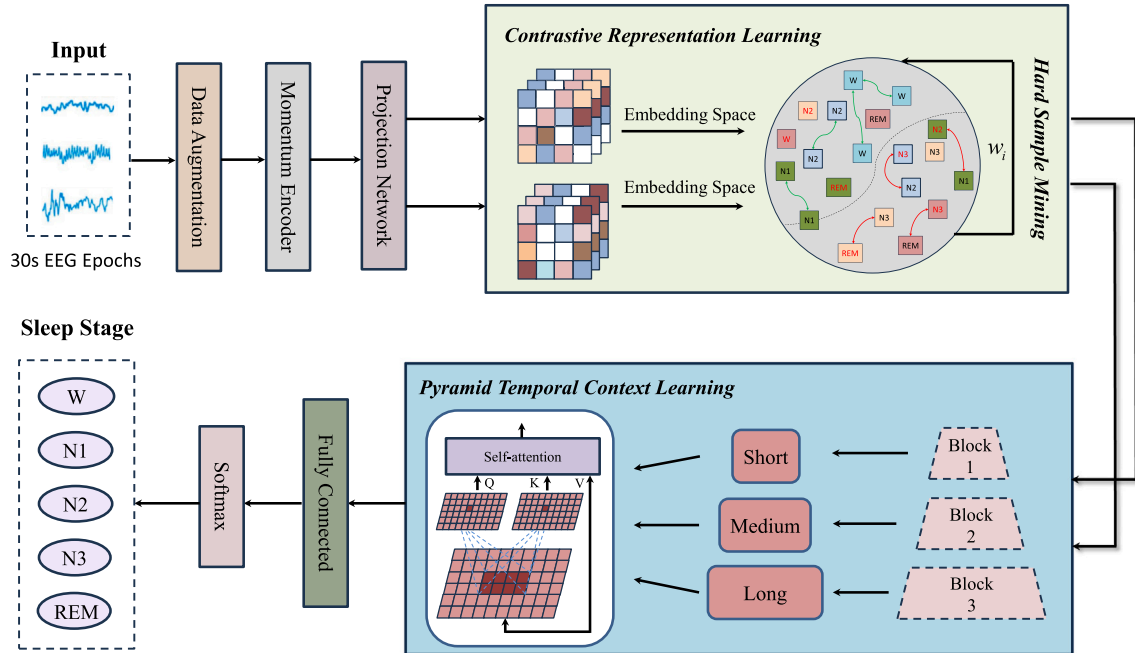


Fig. 1. The overall architecture of the proposed network (MFCR-SleepNet). The architecture comprises two primary components: Contrastive Representation Learning and Pyramid Temporal Context Learning. Initially, the input 30-second EEG segments undergo processing through data augmentation, a momentum encoder, and a projection network, resulting in an embedding space. Within this embedding space, black text represents Positive Samples, while red text denotes Negative Samples. Green lines illustrate the model's process of maximizing similarity between positive samples, whereas red lines indicate the minimization of similarity between negative samples. Through hard sample mining, samples in the embedding space that are difficult to distinguish and lie near boundaries are weighted, further optimizing the contrastive learning objective. Subsequently, the generated features are fed into the Pyramid Temporal Context Learning module, which employs a XSPP-CNN and self-attention mechanisms to perform hierarchical modeling across different time scales (short, medium, long), ultimately predicting the sleep stages.

epochs s_p ($p = 1, \dots, N_b$), the augmented outputs s_q ($q = 1, 2, \dots, 2N_b$) are referred to as augmented sample pairs, where N_b is the batch size (i.e., the number of EEG epochs in one batch).

2) Momentum Encoder: The momentum encoder is designed to extract and maintain stable feature representations of sleep EEG, facilitating contrastive representation learning. The momentum encoder operates on the principle of momentum updating, which allows for smooth updates of network parameters, thereby enhancing the stability and consistency during the learning process.

It takes the augmented EEG signals, denoted as $\tilde{s}$, as inputs and processes them through convolutional layers. This is followed by batch normalization (BN), average pooling, and activation functions to learn more complex and abstract feature representations. It is defined as follows:

$$h_t = \text{ReLU}_{\theta_t}[\text{Avg}\{\text{BN}(\tilde{s})\}] \quad (2)$$

$$\theta_{t+1} = \alpha\theta_t + (1 - \alpha)\theta'_t \quad (3)$$

where θ_t , θ'_t denote the parameters at the current time step, and the updated parameters, both of which are trainable. $\text{Avg}(\cdot)$ and $\text{BN}(\cdot)$ represent the average pooling and batch normalization operations. The momentum coefficient α is set to 0.9 and is used to control the blending ratio between the historical and updated parameters.

3) Projection Network: Direct contrastive learning within the feature space of the encoder might not yield ideal results, especially when the feature space is high-dimensional or lacks sufficient separability. To address this issue, a projection network is employed to map the feature representations extracted by the momentum encoder, denoted as h_t , into an embedding space suitable for contrastive loss computation. The embedding is represented as $z = \frac{z'}{\|z'\|} \in \mathbb{R}^{d_z}$, where $z' = P(h_t; w_p; b_p)$, and $P(\cdot)$ is the projection function, with w_p and b_p representing the weights and biases, respectively. A multilayer perceptron (MLP) [38] with a single hidden layer of dimension $d_z = 256$ is utilized as the projection network.

4) Hard sample mining: Hard sample mining [39] is a widely utilized technique for addressing hard-to-recognize samples in classification tasks, proving particularly advantageous when encountering ambiguous boundaries and imbalanced data distributions. By identifying and concentrating on these challenging samples, the model can allocate more attention to critical instances during training, thereby improving its adaptability in complex scenarios. To effectively mitigate boundary ambiguity and class imbalance in sleep staging, we utilize the characteristics of EEG signals across different sleep stages and propose a hard sample mining approach for the weighted optimization of key samples. This strategy enhances the model's capability to recognize complex instances and improves overall classification performance. Specifically, we concentrate on two main categories of difficult samples: boundary samples and those belonging to classes with low classification accuracy.

Sleep-stage boundary samples typically arise in EEG transition regions, where inherent ambiguity and signal complexity make classification more challenging. Accordingly, we define the boundary sample set based on actual stage-switching points in the ground-truth labels:

$$B = \{i | y_{\text{true}}(i) \neq y_{\text{true}}(i + 1)\} \quad (4)$$

where $y_{\text{true}}(i)$ and $y_{\text{true}}(i + 1)$ denote the ground-truth labels of the i -th and $(i + 1)$ -th time segments.

Different sleep stages display distinct distribution imbalances; for instance, the N1 stage contains relatively few samples with weaker signal characteristics, which typically leads to lower classification accuracy for these categories. Consequently, we classify samples from any class whose accuracy falls below the threshold θ_{accuracy} as hard samples, defined as follows:

$$\mathcal{A} = \{i | y_{\text{true}}(i) = c, \text{Accuracy}(c) < \theta_{\text{accuracy}}\} \quad (5)$$

In our model, θ_{accuracy} is set to 0.7. This value was chosen based on the classification results of the sleep staging task, aiming to better balance the model's classification performance.

By weighting and optimizing these samples, the negative impact of class imbalance on model performance can be significantly reduced. These two categories of hard samples collectively form the target set for hard sample mining:

$$\mathcal{H} = B \cup \mathcal{A} \quad (6)$$

To strengthen the model's focus on hard samples, we integrate a sample weighting mechanism into the Supervised Contrastive Loss (SCL). The weight w_i assigned to each sample is defined as follows:

$$w_i = \begin{cases} \lambda_b, & \text{if } i \in B \\ \lambda_a, & \text{if } i \in \mathcal{A} \\ 1, & \text{other} \end{cases} \quad (7)$$

where λ_a and λ_b are initially set to 1. As training progresses, these parameters are dynamically adjusted according to the number of hard samples and boundary samples. When the contribution of hard samples to the overall loss is minimal, λ_a and λ_b gradually increase to improve the model's focus on these samples. However, when hard samples constitute a large proportion of the data, excessively high weights may cause the loss from hard samples to dominate the total loss, thereby reducing the model's ability to learn from other samples. In more severe cases, this can lead to gradient explosion, causing unstable model training. To prevent instability in the training process due to excessively large weights, we introduce a maximum weight limit w_{max} , restricting the weight values within a reasonable range:

$$w_i = \min(w_i, w_{\text{max}}) \quad (8)$$

where w_{max} is a hyperparameter that limits the maximum weight value. We compared multiple values for w_{max} (1, 1.5, 2, 2.5, 3, 3.5, 4), and the results indicate that setting w_{max} to 3 both ensures that difficult samples receive sufficient attention in the loss computation and avoids overly influencing the overall training process. Therefore, we ultimately set w_{max} to 3 to achieve an effective balance between the impact of difficult samples and the stability of the training process. This approach enables the model to fully focus on hard samples during the early stages of training while gradually stabilizing in later stages, thereby preventing excessive attention to certain samples that could negatively affect overall performance. Additionally, we applied regularization to the updates of λ_a and λ_b , ensuring smooth optimization and preventing drastic fluctuations due to overly rapid updates. This dynamic adjustment mechanism, coupled with the maximum weight constraint, significantly enhances the model's adaptability across different training phases. Consequently, it significantly improves classification performance for minority classes and boundary samples while preserving the model's ability to learn from majority samples and ensuring the stability of the training process.

5) Loss Function: We employ a Supervised Contrastive Loss (SCL) [40] to effectively utilize the embedding vectors derived from the projection network. This loss function maximizes the similarity between positive sample pairs while promoting dissimilarity between negative pairs, thus integrating label information to improve the model's discriminative ability. Specifically, augmented sample pairs with the same annotated sleep stage are treated as positive samples, while pairs from different stages are considered negative samples. Furthermore, we incorporate the previously obtained w_i to apply weighted optimization to hard samples in the SCL, implemented as follows:

$$\mathcal{L}_{\text{sc}} = \sum_{i=1}^{2N_b} \frac{-1}{|N(i)|} \sum_{k=1}^{2N_b} w_i \log \frac{\exp(z_i \cdot z_k / \tau)}{\sum_{a=1}^{2N_b} \exp(z_i \cdot z_a / \tau)} \quad (9)$$

Where z_i and z_k denote different instances of the same sleep stage, $i \neq k$ and $i \neq a$. $N(i)$ denotes the indices of all positive samples in the augmented dataset and $|N(i)|$ denotes the cardinality of this set. τ denotes

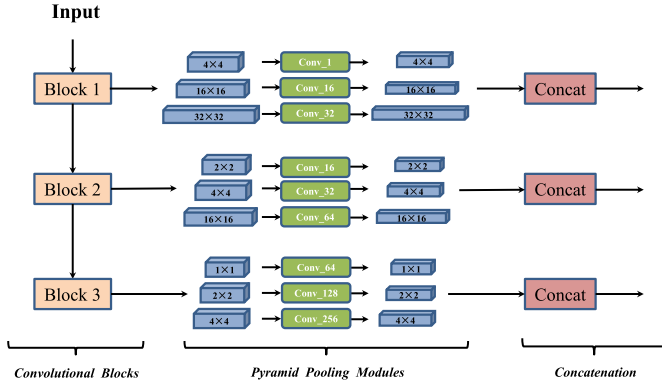


Fig. 2. The framework of XSPP-CNN.

the temperature parameter, set to 0.07 to amplify the effect of small differences between embedding vectors.

2.2. Pyramid temporal context learning

The Pyramid Temporal Context Learning module is designed to capture and synthesize the dynamic characteristics of sleep EEG signals across multiple temporal scales. This approach enables the model to not only understand feature variations within a single EEG epoch but also recognize pattern changes across extended time sequences. This enhances its ability to detect transitions between sleep stages. The module incorporates a self-attention mechanism and a novel network architecture, the Cross-Scale Pyramid Pooling Convolutional Neural Network (XSPP-CNN), which enables effective multi-temporal scale learning of EEG signals.

1) XSPP-CNN: The XSPP-CNN employs a multi-scale convolutional strategy to capture EEG signal features across different temporal ranges, as the characteristics of sleep data typically involve vital physiological signal variations at various temporal scales, which are crucial for accurately identifying different sleep stages. As shown in Fig. 2, the network architecture consists of a multi-scale stack of convolutional layers. Using convolutional kernels of different sizes, it captures a range of contextual information. The network integrates nine convolutional layers into three blocks, followed by pyramid pooling at three scales, enabling the simultaneous capture of short-, medium-, and long-term temporal dependencies. This multi-scale analysis of time-series data significantly enhances the ability to parse complex sleep patterns through cross-scale feature integration.

Initially, the feature vector z learned by CRL is input into the network:

$$z_{out} = PReLU(w_c * z + b_c) \quad (10)$$

where $*$ denotes the convolution operation, with w_c and b_c representing the trainable weights and biases, respectively. To enhance the network's ability to capture subtle differences in the data, PReLU is employed as the activation function.

Subsequently, multi-scale pyramid pooling is performed:

$$r_s = XSPP(z_{out}) \quad (11)$$

Features at different scales are concatenated to capture short-, medium-, and long-term temporal characteristics:

$$x_{concat} = concat(r_1, r_2, \dots, r_s) \quad (12)$$

where r_s denotes the features extracted from the s -th temporal scale during the multi-scale pyramid pooling process.

2) Self-Attention Mechanism: In the self-attention mechanism, after extracting multi-scale temporal features from the XSPP-CNN, feature

vectors corresponding to short, medium, and long temporal scales are standardized in length through padding before being fed into the mechanism. This mechanism effectively calculates long-range dependencies within the data, enabling it to identify and highlight key information by computing attention scores across different temporal points, thereby filtering out irrelevant details. It is defined as follows:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (13)$$

where Q , K , V denote Query, Key and Value, which are generated from the input x_{concat} using three distinct trainable matrices W^Q , W^K , and W^V , respectively.

Finally, the output probabilities of the sleep stages, denoted as p , are obtained through a fully connected layer and evaluated using cross-entropy loss to assess classification accuracy:

$$\mathcal{L}_{ce} = - \sum_{c=1}^C y_{o,c} \log(p_{o,c}) \quad (14)$$

where $y_{o,c}$ is a binary indicator that represents whether the observed sample o belongs to category c ; $p_{o,c}$ denotes the probability predicted by the model that sample o belongs to category c , C denotes the total number of categories in the classification task, which corresponds to the five stages of sleep, thus $C = 5$.

3. Experiments

In this section, we introduce the three public datasets used to validate our proposed model and provide a detailed description of our experimental procedure.

3.1. Dataset

In this study, we use three publicly available datasets: SleepEDF-78, MASS, and SHHS—and perform relevant preprocessing to validate the effectiveness of our MFCR-SleepNet model.

1) SleepEDF-78: The SleepEDF-78 database is the 2018 version of the Sleep-EDF Expanded Dataset [41,42], containing data from 78 healthy subjects aged 25–101. Each PSG record includes two scalp EEG signals (Fpz-Cz and Pz-Cz), one horizontal EOG channel, one EMG channel, and one nasal/oral airflow channel. All EEG and EOG signals are sampled at 100 Hz. The collected data were manually annotated by sleep experts using the R&K standard [9], with each 30-second PSG segment labeled as one of eight categories: W, N1, N2, N3, N4, REM, MOVEMENT, UNKNOWN. In this study, we adopted the raw EEG signals from the Fpz-Cz channel.

2) Montreal Archive of Sleep Studies (MASS): MASS [43] contains polysomnographic recordings from 200 subjects (97 males and 103 females), with ages ranging from 18 to 76 years. The dataset is divided into five subsets (SS1 to SS5), classified according to different research purposes and data acquisition protocols. Subsets SS1 and SS3 are annotated using the AASM standard, while SS2, SS4, and SS5 follow the R&K standard. The SS3 subset includes recordings from 62 healthy subjects, with each recording containing EEG, EOG, EMG, and ECG channels, sampled at 256 Hz. Since the other methods we compared utilized the SS3 subset, we adopted the same data selection strategy and used the C4 EEG channel to ensure a fair SOTA comparison with these models.

3) Sleep Heart Health Study (SHHS): SHHS [44] is a multicenter cohort study designed to explore the impact of sleep apnea on cardiovascular diseases. The dataset includes two rounds of polysomnographic (PSG) recordings: Visit 1 (SHHS-1) and Visit 2 (SHHS-2). The SHHS-1 dataset contains recordings from 5,793 subjects aged 39 to 90 years. Each recording includes two-channel EEG, two-channel EOG, single-channel EMG, single-channel ECG, two-channel respiratory inductance plethysmography, as well as data from position sensors, light sensors, pulse oximeters, and airflow sensors. Sleep stages are manually labeled

as W, REM, N1, N2, N3, N4, MOVEMENT, and UNKNOWN according to the R&K rules. In this study, we analyzed single-channel EEG signals from the C4-A1 channel in 5,793 SHHS-1 recordings.

3.2. Data preprocessing

We performed several preprocessing steps on the three datasets. First, we extracted raw EEG signals from the relevant channels, then applied standard filtering techniques to remove ocular and muscular artifacts. Next, we divided the data into 30-second, non-overlapping segments. In all three datasets, we merged N3 and N4 stages into a single N3 stage following AASM standards, and excluded segments labeled as “MOVEMENT” or “UNKNOWN,” as these are unsuitable for sleep stage classification [10]. Finally, all signals were uniformly resampled to 100 Hz to ensure consistency across the datasets.

3.3. Training

In the training phase, we employed 10-fold cross-validation to evaluate the performance of our proposed model. The model was optimized using the Adam optimizer with hyperparameter settings $\eta = 1 \times 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.99$ and $\epsilon = 1 \times 10^{-8}$. L2 regularization was applied to prevent overfitting. The initial learning rate was set to 0.0005, with a batch size of 512. To further improve generalization and prevent overfitting, we adopted an early-stopping strategy: if the validation loss did not decrease for 20 consecutive validation steps, training was terminated automatically. At each cross-validation round, we evaluated the model with the lowest validation loss on the test set. All model development and training procedures were implemented in Python 3.10.2 using an Nvidia GeForce GTX 3090 GPU.

3.4. Evaluation indicators

Several standard metrics were used to evaluate the overall performance of the model in this study. Accuracy (ACC) measures the ratio of correctly predicted samples to the total number of samples, and is formulated as:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

The macro-averaged F1 score (MF1) combines precision and recall, making it particularly useful for multi-class tasks with imbalanced data. It provides a comprehensive reflection of model performance and is expressed as:

$$MF1 = \frac{1}{C} \sum_{i=1}^C \frac{2 * Precision * Recall}{Precision + Recall} \quad (16)$$

where $Precision = \frac{TP}{TP+FP}$, $Recall = \frac{TP}{TP+FN}$, and C denotes the number of sleep-stage classes.

Cohen's Kappa is used to measure the agreement between the model's predictions and the ground-truth labels, correcting for chance agreement. The per-class F1 score reflects the model's ability to distinguish each individual class.

The Boundary Error Rate (BER) is a newly proposed metric designed to address a recognized gap in sleep stage classification evaluation. Recent surveys on time series segmentation have noted that existing state detection methods “do not attach sufficient importance to accurate detection of boundary points in evaluation” [45]. Traditional metrics such as accuracy and F1 scores may mask misclassifications at transitional stages, which are clinically relevant [46]. BER integrates two complementary dimensions. The Temporal Error (TE) component adopts the mean absolute error approach commonly used in change point detection, measuring the temporal distance between predicted and true

boundaries. The Phase Consistency (PC) component evaluates the correctness of post-transition stage labels, reflecting the clinical importance of sleep stage transitions [47]. The metric is defined as follows:

$$TE = \frac{1}{N} \sum_{i=1}^N |t_{true,i} - t_{pred,i}| \quad (17)$$

Where $t_{true,i}$ denotes the position of the i -th true boundary, $t_{pred,i}$ is the predicted boundary closest to that true boundary, and N denotes the total number of true boundaries. A smaller boundary error indicates greater accuracy in boundary localization. To normalize TE, we define: $TE_{norm} = \frac{TE}{TE_{max}}$. Where TE_{max} is the maximum observed error. The formula for PC is given as follows:

$$PC = \begin{cases} 1, & \text{if } y_{pred,t} = y_{true,t} \\ 0, & \text{otherwise} \end{cases} \quad (18)$$

Where $y_{pred,t}$ denotes the model-predicted stage label immediately following boundary t , and $y_{true,t}$ denotes the true stage label after the same boundary. We then compute the average phase consistency PC as $PC_{avg} = \frac{1}{N} \sum_{i=1}^N PC(t_{true,i})$.

By applying weighted multiplication to TE and PC_{max} , we obtain the final integrated metric, BER. The formula is expressed as follows:

$$BER = \alpha \cdot TE_{norm} + \beta (1 - PC_{avg}) \quad (19)$$

where α and β are the weighting factors, each set to 0.5 in this study. The term $(1 - PC_{avg})$ serves as an inverse indicator of phase consistency, because better phase consistency lowers the BER value. In other words, a smaller BER indicates better model performance in boundary classification. As no other models have evaluated this metric, we calculated it only for the baseline network and compared it with our model.

3.5. Baseline networks

In this experiment, we compare our proposed method with several well-established baseline networks, including DeepSleepNet [29], XSleepNet [19], and SeqSleepNet [27]. A brief overview of these networks is as follows:

- **DeepSleepNet [29]**: DeepSleepNet uses a Convolutional Neural Network (CNN) to extract time-invariant features and employs a Bidirectional Long Short-Term Memory (BiLSTM) network to automatically learn the transition patterns between sleep stages. This model adopts a two-step training strategy that addresses the issue of automatic feature extraction from single-channel EEG signals.
- **XSleepNet [19]**: XSleepNet is a multi-view sequence-to-sequence sleep staging model that combines raw signals with time–frequency images. By dynamically adjusting the learning rates across different views, the model refines a joint representation. XSleepNet mitigates the issue of insufficient complementarity between multiple modalities.
- **SeqSleepNet [27]**: SeqSleepNet is based on a hierarchical recurrent neural network (RNN). Through frequency-domain feature extraction and short- and long-term sequence modeling, it can simultaneously predict labels from multiple consecutive sleep segments. This model extends the sleep staging task from independent classification to sequence classification, thereby capturing dynamic relationships among stages.

To ensure an accurate and fair comparison of model performance, we performed 10-fold cross-validation on each dataset for every model, employing a leave-one-out approach in each fold. The training strategy strictly follows the settings described in the “Training” section, thus ensuring fairness and comparability in the evaluation results.

4. Results and discussion

4.1. Confusion matrix

Fig. 3 shows the confusion matrices of the model on the SleepEDF-78, MASS, and SHHS datasets, highlighting its classification performance for each sleep stage. The classification accuracies for the W, N2, N3, and REM stages all exceed 89 %, demonstrating the model's robust classification capabilities for major sleep stages. Notably, the accuracy for the N1 stage reaches 57 % (SleepEDF-78), 55 % (MASS), and 52 % (SHHS), representing a significant improvement over traditional methods. Specifically, the introduction of a hard sample mining strategy effectively mitigates the challenges of insufficient samples and ambiguous boundaries in the N1 stage, providing the model with a clear performance advantage over existing approaches in this transitional stage. These results indicate that the model not only excels in

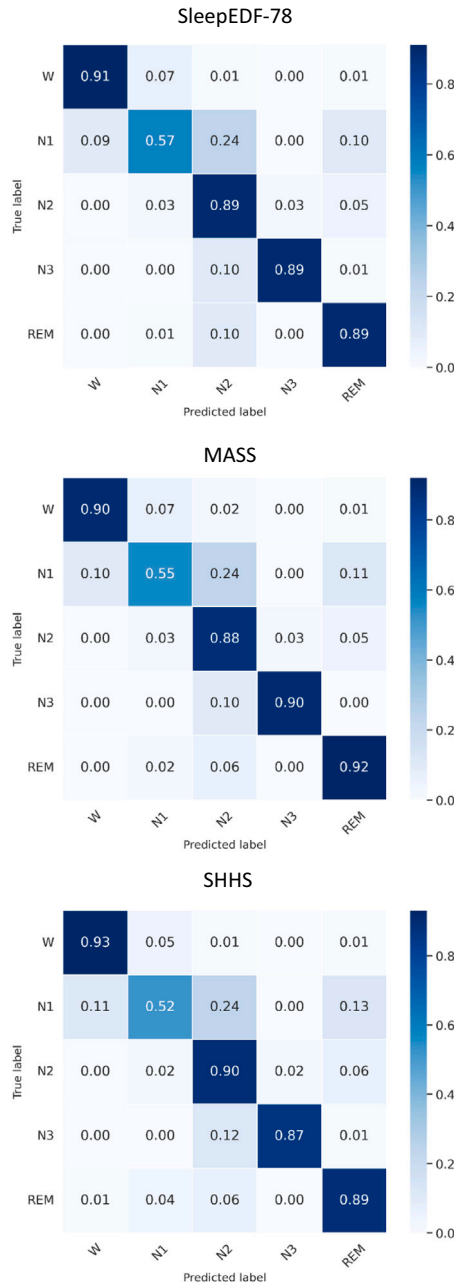


Fig. 3. Confusion matrices on three datasets.

classifying the primary stages but also significantly enhances its discriminative power for more challenging samples.

4.2. Sleep hypnogram

Fig. 4 compares the model's predicted sleep stages (lower panel) with the ground-truth labels (upper panel), where red dots indicate misclassified segments. The results demonstrate that the model accurately captures the transition trends of major sleep stages (e.g., W, N2, N3, and REM) and maintains high consistency with the true labels, particularly exhibiting strong classification performance in stable stages such as N2 and N3. However, misclassifications are predominantly concentrated in transitional stages (e.g., between N1 and N2, and between N2 and N3) due to the significant overlap and ambiguity in the signal characteristics of these stages.

By incorporating a hard sample mining strategy, our approach effectively enhances the model's robustness in handling boundary samples and complex transitional phases. Specifically, the number of errors in transitional stages is markedly reduced compared to traditional methods, allowing the model to better capture subtle signal variations in boundary regions. This improvement is reflected in the distribution of red dots in the figure, where errors, although concentrated near boundaries, are fewer and more evenly distributed. Furthermore, in extended continuous segments of major stages (e.g., N2 and REM), virtually no significant errors are observed, further validating the model's ability to extract features from stable stages.

4.3. Compared with state-of-the-art approaches

To validate the effectiveness of our proposed method, we compared it with several state-of-the-art sleep staging models—Pradeepkumar et al. [48], U-Time [49], SAN [50], CTCNet [51], SleepSatelightFTC [52] and SleepPyCo [23]—as well as baseline models (DeepSleepNet [29], SeqSleepNet [27], and XSleepNet [19]) on three public datasets: SleepEDF-78, MASS, and SHHS. Table 1 presents a comparison of overall metrics (accuracy, macro-averaged F1 score, and Cohen's Kappa), as well as per-stage F1 scores.

Compared to other state-of-the-art methods, our algorithm demonstrates outstanding performance on both SleepEDF-78 and MASS, achieving accuracies of 85.9 % and 87 %, respectively. This highlights the method's efficiency and accuracy in handling diverse types of sleep data. Specifically, for the MASS dataset—encompassing more complex and diverse sleep patterns—our model's high accuracy highlights its excellent feature extraction and classification capabilities. On the SHHS dataset, our model's accuracy of 87.9 % is comparable to that of SleepPyCo, which is specifically designed to improve recognition accuracy for the challenging-to-distinguish N1 and REM stages. This suggests that our model performs similarly in capturing subtle features of these two stages, where overlapping characteristics and ambiguous boundaries are common.

The examination of the per-stage F1 scores reveals that the N1 stage is often considered the most challenging to classify due to its weak signal features and the inherent boundary ambiguity resulting from its transitional nature. Our model's classification accuracy for the N1 stage significantly outperforms that of existing models, such as U-Time [49], DeepSleep [29], and SleepPyCo. For instance, on the SleepEDF-78 dataset, our F1 score reaches 55.1 %, surpassing those of other methods. This can be attributed to the integration of a hard sample mining strategy and a supervised contrastive learning approach, which enable the model to effectively identify and handle challenging boundary samples.

4.4. Interpretability analysis of attention mechanisms

To enhance the interpretability of the model [53], we conducted a visualization analysis of self-attention weights. Fig. 5 illustrates the attention distributions for three representative sleep stages (N1, N2, and

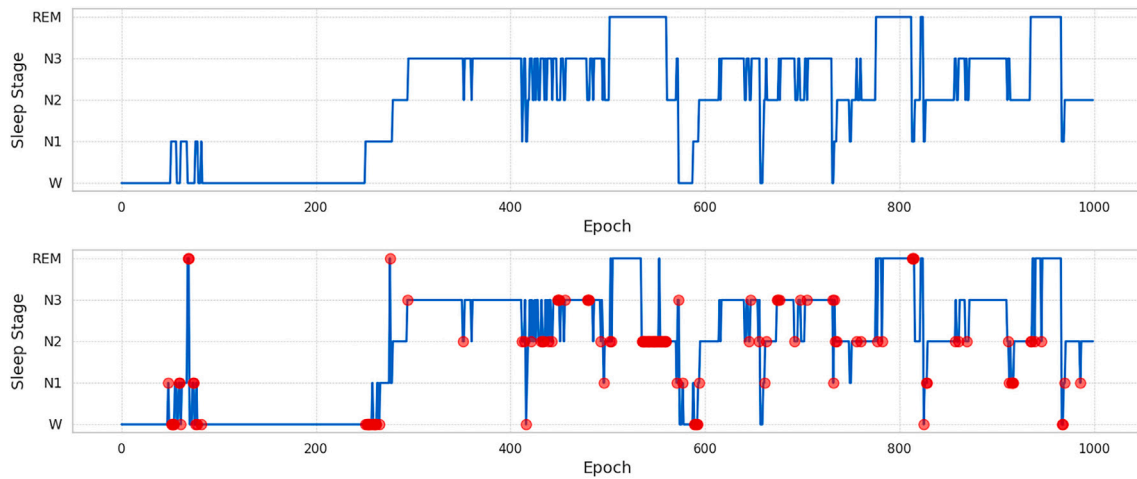


Fig. 4. Visualization of sleep stages for the random 1000 epochs of one subject in SleepEDF-78. (a) The ground truth hypnogram; (b) the output hypnogram, where red dots indicate misclassified epochs.

Table 1
Comparison with state-of-the-art approaches on the three datasets.

Database	Method		Overall results			Pre class F1-score				
	Model	Channel	ACC	MF1	Kappa	W	N1	N2	N3	REM
SleepEDF-78	Ours	Single-Channel	85.9	79.7	0.789	93.7	55.1	86.9	86.0	86.7
	DeepSleepNet [29]	Single-Channel	82.0	76.9	0.76	84.7	46.6	85.9	84.8	82.4
	SeqSleepNet [27]	Multi-Channel	82.6	76.4	0.76	-	-	-	-	-
	XSleepNet [19]	Multi-Channel	84.0	77.9	0.778	-	-	-	-	-
	SleepPyCo [23]	Single-Channel	84.6	79.0	0.787	93.5	50.4	86.5	80.5	<u>84.2</u>
	Pradeepkumar et al [48]	Multi-Channel	84.3	79.4	0.781	93.5	<u>54.3</u>	85.5	77	86.7
	U-Time [49]	Single-Channel	81.3	76.3	0.745	92.0	51.0	83.5	74.6	80.2
	SAN [50]	Single-Channel	<u>85.8</u>	77.6	0.854	94.5	37	84.5	96.8	84.1
	CTCNet [51]	Single-Channel	82.5	79.1	0.78	92.5	53.8	<u>86.8</u>	<u>87.3</u>	74.8
	SleepSatelightFTC [52]	Single-Channel	84.8	77.8	0.787	<u>93.8</u>	47.4	86.5	79.5	82.1
MASS	Ours	Single-Channel	87.0	<u>82.2</u>	<u>0.836</u>	<u>91.4</u>	61.8	90.7	84	91.2
	DeepSleepNet [29]	Single-Channel	86.2	81.7	0.8	87.3	59.8	90.3	81.5	89.3
	SeqSleepNet [27]	Multi-Channel	84.5	79.8	0.778	-	-	-	-	-
	XSleepNet [19]	Multi-Channel	85.2	80.6	0.788	-	-	-	-	-
	SleepPyCo [23]	Single-Channel	<u>86.8</u>	82.5	0.811	89.2	<u>60.1</u>	90.4	<u>83.8</u>	89.1
	SAN [50]	Single-Channel	86.4	80.3	0.859	98.9	40.3	<u>90.5</u>	81.0	<u>90.7</u>
SHHS	Ours	Single-Channel	87.9	81.8	0.835	93.1	54.2	<u>89.2</u>	<u>85.0</u>	<u>89.4</u>
	SeqSleepNet [27]	Multi-Channel	86.5	78.5	0.81	-	-	-	-	-
	XSleepNet [19]	Multi-Channel	87.6	80.7	0.826	-	-	-	-	-
	SleepPyCo [23]	Single-Channel	87.9	80.7	<u>0.83</u>	92.6	49.2	88.5	84.5	88.6
	Pradeepkumar et al [48]	Multi-Channel	<u>87.7</u>	<u>81.4</u>	0.829	<u>92.8</u>	<u>52.5</u>	88.3	83.7	89.8
	CTCNet [51]	Single-Channel	85.7	77.2	0.81	87.1	33.6	89.5	90.9	83.5

REM). Each sample comprises 10 consecutive 30-second epochs (totaling 300 s), with the model predicting the label of the final epoch. Our analysis reveals three distinct attention patterns corresponding to stages with varying feature intensities, highlighting the adaptive behavior of the self-attention mechanism.

Specifically, for the N1 stage (Fig. 5(a)), the attention weights are predominantly concentrated at the end of the sequence (Max: 0.0083), exhibiting a "current state prioritization" pattern. This aligns with the characteristics of N1 as a transitional stage with subtle signal features, and it explains why hard sample mining strategies significantly improve N1 performance. For the N2 stage (Fig. 5(b)), multiple local peaks (Max: 0.0144) are observed, which correlate temporally with transient feature events such as sleep spindles and K-complexes, indicating the model's ability to automatically localize discriminative events. For the REM stage (Fig. 5(c)), the highest attention peak (Max: 0.0150) is sharply focused on the final epoch, demonstrating that the model successfully captures the distinctive features of REM, enabling high-confidence predictions.

The maximum attention weights across the three stages show an increasing trend (0.0083 $\rightarrow$ 0.0144 $\rightarrow$ 0.0150), reflecting the model's

adaptive sensitivity to the distinctiveness of features in different stages: conservative attention for ambiguous transitional stages, selective focus on event-based stages, and high-confidence attention for stages with clear features. This adaptive behavior closely resembles the scoring logic of human sleep experts, providing support for the clinical reliability of deep learning-based sleep stage classification systems.

4.5. Boundary error rate

The Boundary Error Rate (BER) is an indicator used to evaluate a model's accuracy in detecting transitions between different sleep stages during a sleep cycle. BER integrates the Temporal Error (TE) at the boundary and the Phase Consistency (PC) following the boundary; the lower the BER, the more accurately the model identifies stage transitions. In this study, we compare the BER performance of our method with three baseline networks—DeepSleepNet, SeqSleepNet, and XSleepNet—on the SleepEDF-78, MASS, and SHHS datasets. As shown in Fig. 6, our approach demonstrates clear advantages across all datasets.

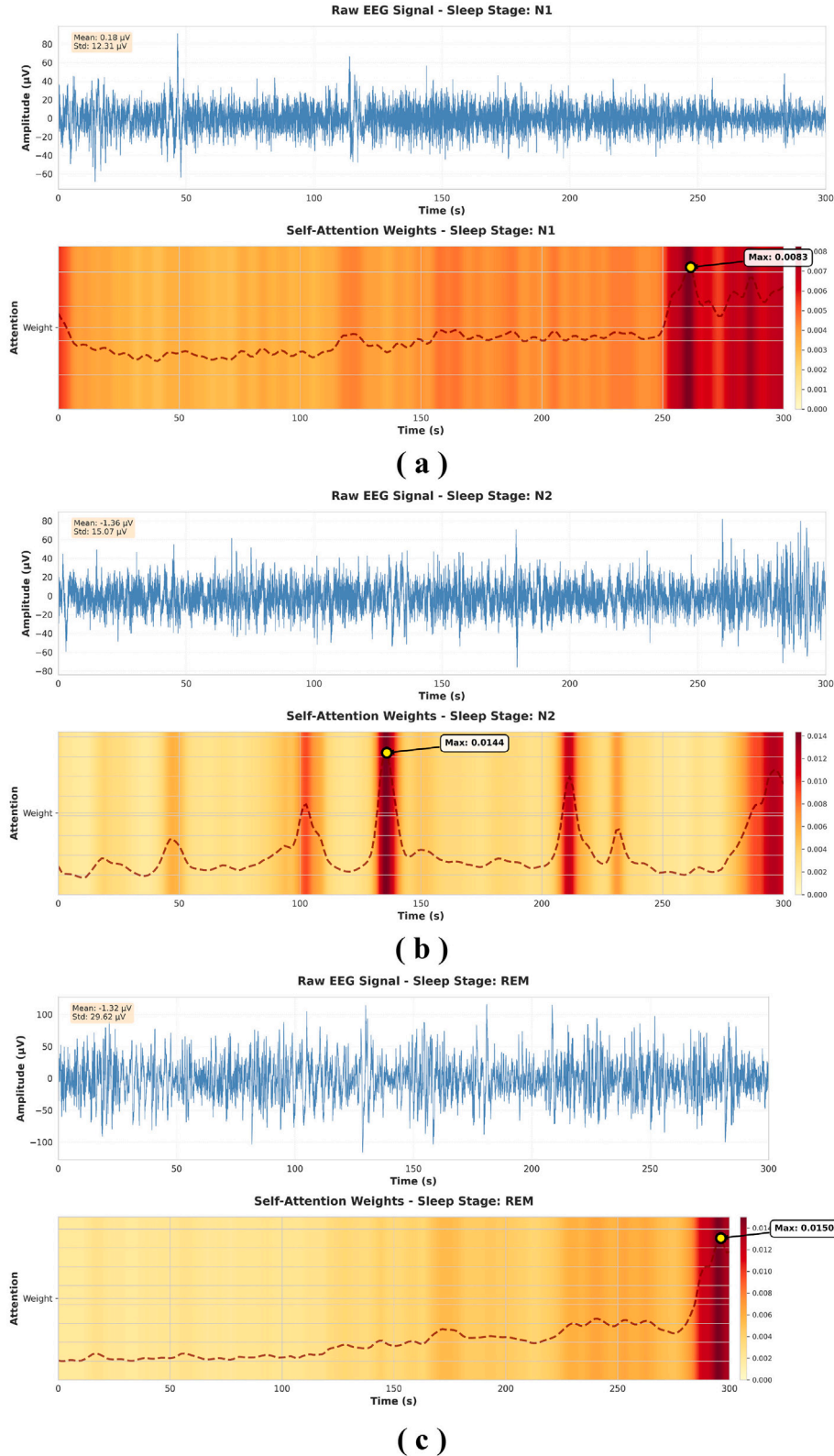


Fig. 5. Self-attention weight visualization for three representative sleep stages. Each panel shows raw EEG signal (upper, 300s sequence of 10 $\times$ 30s epochs) and attention heatmap (lower) when predicting the final epoch. (a) N1: concentrated on recent epochs (Max: 0.0083). (b) N2: multiple peaks aligned with transient events (Max: 0.0144). (c) REM: highest peak on final epoch (Max: 0.0150). Progressive increase in maximum weights reflects adaptive perception of feature saliency.

Specifically, on the SleepEDF-78 dataset, our model achieves a BER of 0.2174, significantly outperforming DeepSleepNet (0.2595), SeqSleepNet (0.2941), and XSleepNet (0.2463). This demonstrates the effectiveness of our method in accurately detecting sleep stage transition

points. On the MASS dataset, our model also excels, achieving a BER of 0.1932 and surpassing all other models. This result highlights the adaptability and strengths of our model when handling more complex data. Finally, on the larger SHHS dataset, despite its higher complexity and

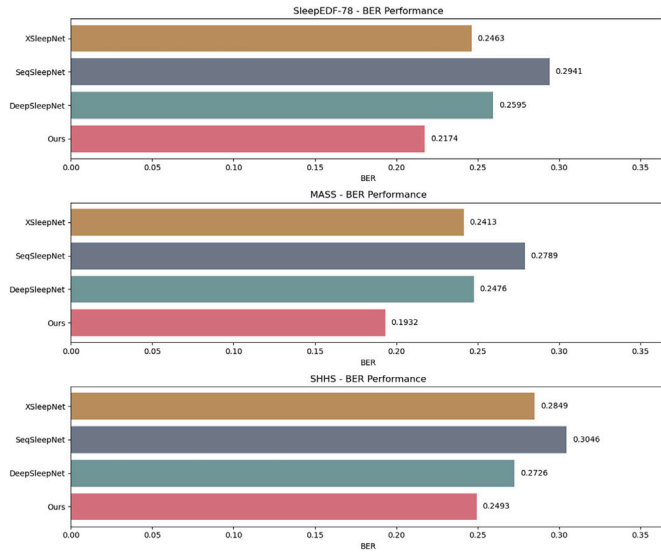


Fig. 6. Boundary error rate (BER) performance across three datasets.

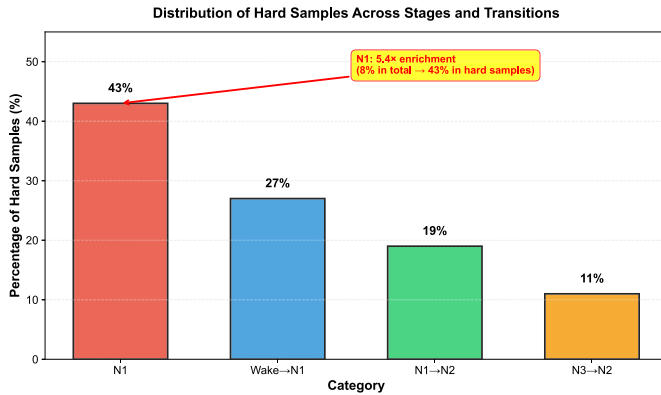


Fig. 7. Hard sample distribution across sleep stages and transitions.

challenges, our method still demonstrates robust performance, achieving a BER of 0.2493, lower than that of DeepSleepNet (0.2726), SeqSleepNet (0.3046), and XSleepNet (0.2849).

4.6. Hard sample analysis

To gain deeper insight into the mechanisms underlying BER improvement, we further analyze the model's performance on hard samples. Hard samples, defined as epochs with classification accuracy below 70 % or located at stage transition boundaries, account for 19.3 % of the test set. As shown in Fig. 7, hard samples are predominantly concentrated in the N1 stage (43 % of hard samples, despite N1 comprising only 8 % of the overall test set—a 5.4-fold enrichment) and at stage transitions (Wake→N1: 27 %, N1→N2: 19 %). This distribution pattern is highly consistent with clinical observations: literature reports indicate that inter-rater agreement among expert scorers is significantly lower for the N1 stage and transition boundaries (Cohen's $\kappa < 0.6$) compared to other stages ($\kappa > 0.8$) [54], demonstrating that classification difficulty in these regions represents an inherent challenge in sleep staging tasks.

Our model achieves approximately 64 % accuracy on hard samples, significantly outperforming the baseline method DeepSleepNet's approximately 52 % ($p < 0.01$, McNemar's test). Notably, this improvement margin exceeds that of overall accuracy, indicating that our approach is particularly effective in handling challenging samples. Further analysis reveals that for the two most challenging sample categories, N1

Table 2

Ablation Experiments on SleepEDF-78.

Method	Overall results			Pre class F1-score				
	ACC	MF1	Kappa	W	N1	N2	N3	REM
EN + AP	81.0	74.8	0.741	90.7	42.5	83.1	75.0	76.3
EN + MSP	83.3	76.4	0.759	91.9	46.3	84.5	77.8	79.6
ME + AP	82.9	76.1	0.752	91.6	46.7	83.9	77.1	78.4
ME + MSP	83.7	77.7	0.773	93.4	48.1	85.3	80.5	81.4
ME + MSP + HSM	85.9	79.7	0.789	93.7	55.1	86.9	86.0	86.7

stage hard samples improve from approximately 39 % to 52 % accuracy, while Wake→N1 transition boundaries improve from approximately 45 % to 62 %.

These findings provide mechanistic explanations for the BER improvements observed in Section 4.5. Since stage transitions constitute the primary source of hard samples, and our method demonstrates significant performance gains precisely in these regions, the reduction in BER directly reflects the model's enhanced capability in handling transition boundaries. The synergistic effect of hard sample reweighting and pyramid temporal context learning enables the model to develop more robust representations for challenging samples—particularly at stage boundaries where multiple physiological features overlap and evolve dynamically. This enhanced robustness on hard samples, especially at stage transitions, directly translates to lower BER, as these boundary samples are the primary source of temporal errors in sleep staging systems.

4.7. Ablation experiments

In our ablation experiments, we assessed the contribution of each component in MFCR-SleepNet by comparing five different configurations: EN + AP (encoding network with average pooling), EN + MSP (encoding network with pyramid pooling), ME + AP (momentum encoder with average pooling), ME + MSP (momentum encoder with pyramid pooling), and ME + MSP + HSM (our method, integrating a momentum encoder, pyramid pooling, and hard sample mining). As shown in Table 2, the results indicate that our method (ME + MSP + HSM) outperforms the other configurations in terms of overall accuracy (ACC), macro-averaged F1 score (MF1), and Cohen's Kappa, validating the effectiveness of the momentum encoder, multi-scale pyramid pooling, and hard sample mining.

Specifically, the introduction of the momentum encoder significantly enhances the stability of feature representations by effectively reducing fluctuations during the learning process through its continuous updating mechanism, which is essential for distinguishing similar sleep stages. Meanwhile, the multi-scale pyramid pooling strategy extracts features at different temporal scales, further enhancing the model's ability to capture dynamic variations in EEG signals. Moreover, the application of hard sample mining (HSM) significantly improves the model's performance in recognizing boundary-ambiguous regions and difficult-to-classify stages.

To further validate the effectiveness of feature representations for each configuration, we present two t-SNE comparison plots in Fig. 8. Plot (a) shows the feature distribution under the ME + MSP configuration, where it can be observed that although different sleep stages exhibit some separation in the feature space, the boundaries between certain classes remain somewhat ambiguous, particularly with poor clustering performance for the N1 category. To further analyze this, we zoomed in on the N1 region to observe its feature distribution in more detail. This adjustment highlights the challenges the model faces in distinguishing similar sleep stages. Plot (b) shows the feature distribution for the ME + MSP + HSM configuration. In comparison, it is evident that after incorporating hard sample mining, the separation between classes in the feature space becomes clearer, and the clusters tighter. This indicates that HSM not only improves the model's overall performance but also strengthens the discriminative ability between different sleep stages at the feature level.

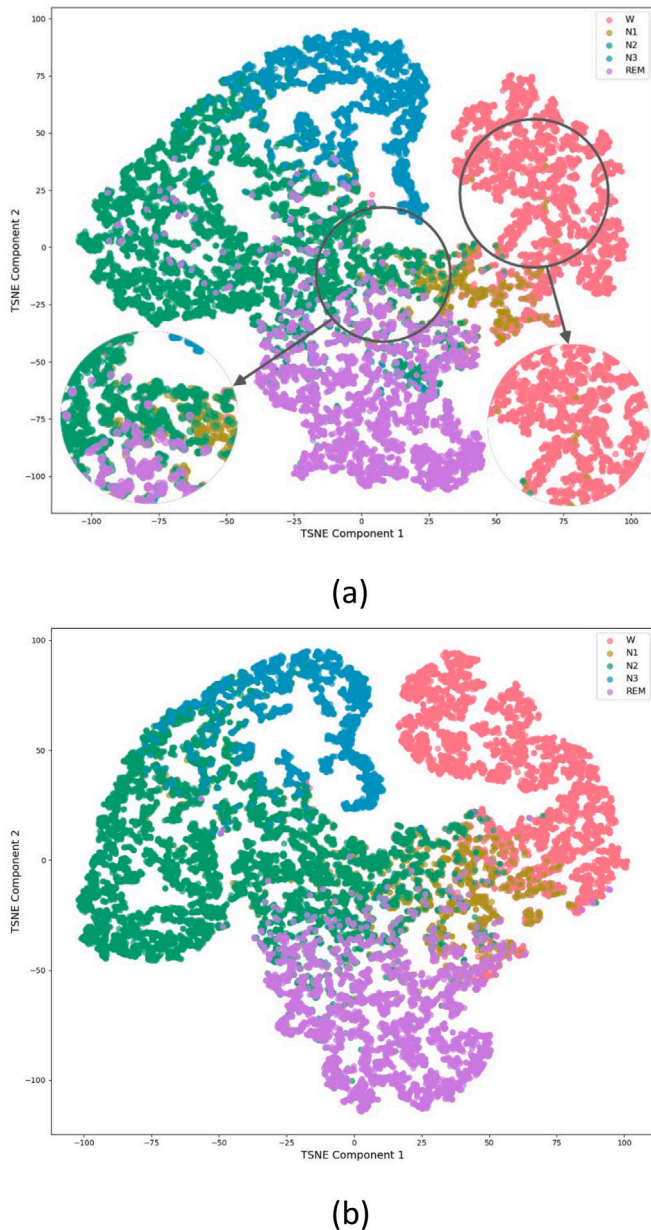


Fig. 8. t-SNE Plots from ablation study on SleepEDF-78 (a) ME + MSP; (b) ME + MSP + HSM.

4.8. Performance-efficiency comparison analysis of single-channel and multi-channel methods

To evaluate the practical performance boundaries of single-channel methods, we conducted a systematic comparison between our proposed method and multi-channel methods reported in the literature (Table 1). On the Sleep-EDF-78 dataset, our single-channel method achieved an accuracy of 85.9 %, surpassing multi-channel methods such as SeqSleepNet (82.6 %), XSleepNet (84.0 %), and Pradeepkumar et al. (84.3 %) by 3.3, 1.9, and 1.6 percentage points, respectively. Similar trends were observed on the MASS and SHHS datasets. These results suggest that, with appropriate architectural design, single-channel methods can achieve performance levels comparable to or slightly exceeding those of multi-channel methods. We attribute this to the synergistic effect of hard sample mining and pyramidal temporal context learning, which enables the model to extract more discriminative representations from limited single-channel inputs, partially compensating for the reduced input information.

While achieving comparable or slightly superior performance, the single-channel method offers significant resource advantages across multiple dimensions. From a hardware perspective, portable single-channel EEG devices are considerably less costly than standard polysomnography (PSG) systems. From a data transmission perspective, single-channel data volumes are smaller than those of multi-channel PSG, which is particularly advantageous for telemedicine and cloud-based transmission scenarios. From a computational efficiency perspective, our model has significantly fewer parameters than typical multi-channel sequence models, resulting in shorter inference times and better suitability for edge device deployment. From a patient compliance perspective, simplified single-channel devices offer greater comfort, leading to higher acceptance for long-term use [55]. These advantages make single-channel methods more suitable for long-term home monitoring, large-scale population screening, edge device deployment, and resource-constrained environments.

Based on the above analysis, we conclude that single-channel and multi-channel methods each have their applicable scenarios. For scenarios requiring the establishment of diagnostic gold standards or detailed analysis of sleep physiology, the rich physiological information provided by multi-channel methods remains irreplaceable. However, for routine sleep stage classification, long-term monitoring, large-scale screening, and resource-constrained environments, well-designed single-channel methods can deliver satisfactory performance while maintaining resource efficiency, thereby offering greater accessibility and sustainability. The contribution of this study lies in demonstrating the potential to enhance performance under single-channel constraints through architectural innovation.

5. Conclusion

In this work, we tackle two major challenges in sleep stage classification: boundary ambiguity between adjacent stages and class imbalance. We propose a Multiscale Feature Contrastive Representation Learning Sleep Network (MFCR-SleepNet), which integrates supervised contrastive learning, hard sample mining, and pyramid temporal context learning. This approach comprehensively captures the dynamic characteristics of sleep EEG signals at short-, medium-, and long-term scales while effectively improving classification performance for minority stages and boundary-transition stages. Moreover, the newly introduced Boundary Error Rate (BER) provides a multidimensional evaluation framework for assessing predictive accuracy at sleep-stage switching points. Experimental results on three public datasets—SleepEDF-78, MASS, and SHHS—demonstrate that MFCR-SleepNet achieves superior accuracy and robustness, outperforming several mainstream models. As a newly proposed metric, BER's clinical utility warrants further validation through independent clinical studies. This study offers new insights into enhancing the reliability and accuracy of automated sleep staging and lays a solid foundation for the diagnosis and intervention of sleep disorders.

CRediT authorship contribution statement

Bohan Li: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Songyun Xie:** Project administration, Methodology, Funding acquisition. **Xinzhou Xie:** Formal analysis, Conceptualization. **Hao Tang:** Validation, Supervision, Project administration. **Michael Scholz:** Validation, Software, Data curation. **Zhongyu Tian:** Software, Data curation. **Zhili Zhang:** Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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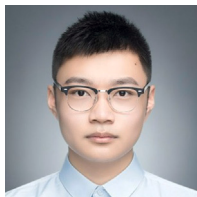
Data availability

Data will be made available on request.

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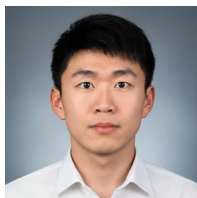
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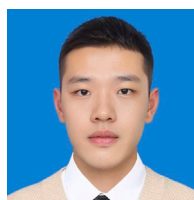
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